

Human vs LLM Text Classification Using a Multi-Feature Approach



Hyungbin Park¹, Kihoon Lee², Hanwool Kim¹, Cheoljun Kim¹, Shinsung Kang¹ and Gwangsu Kim^{1,2}

¹ Department of Statistics, College of Natural Sciences, Jeonbuk National University

² Research Institute for Materials and Energy Sciences, Jeonbuk National University

Introduction

Key question: To what extent can the fundamental differences between human and LLM texts be quantified through a measurable set of linguistic features?

Advanced Large Language Models (LLMs), now generate text that is nearly indistinguishable from human writing. This creates practical problems in areas requiring source verification, such as ensuring academic integrity and preventing the spread of AI-generated spam. Our research develops a classification method utilizing various linguistic features such as lexical density, Personal Pronoun Proportion (PRP), and thematic features. We expect that a more accurate distinction is possible by combining multiple types of linguistic features simultaneously.

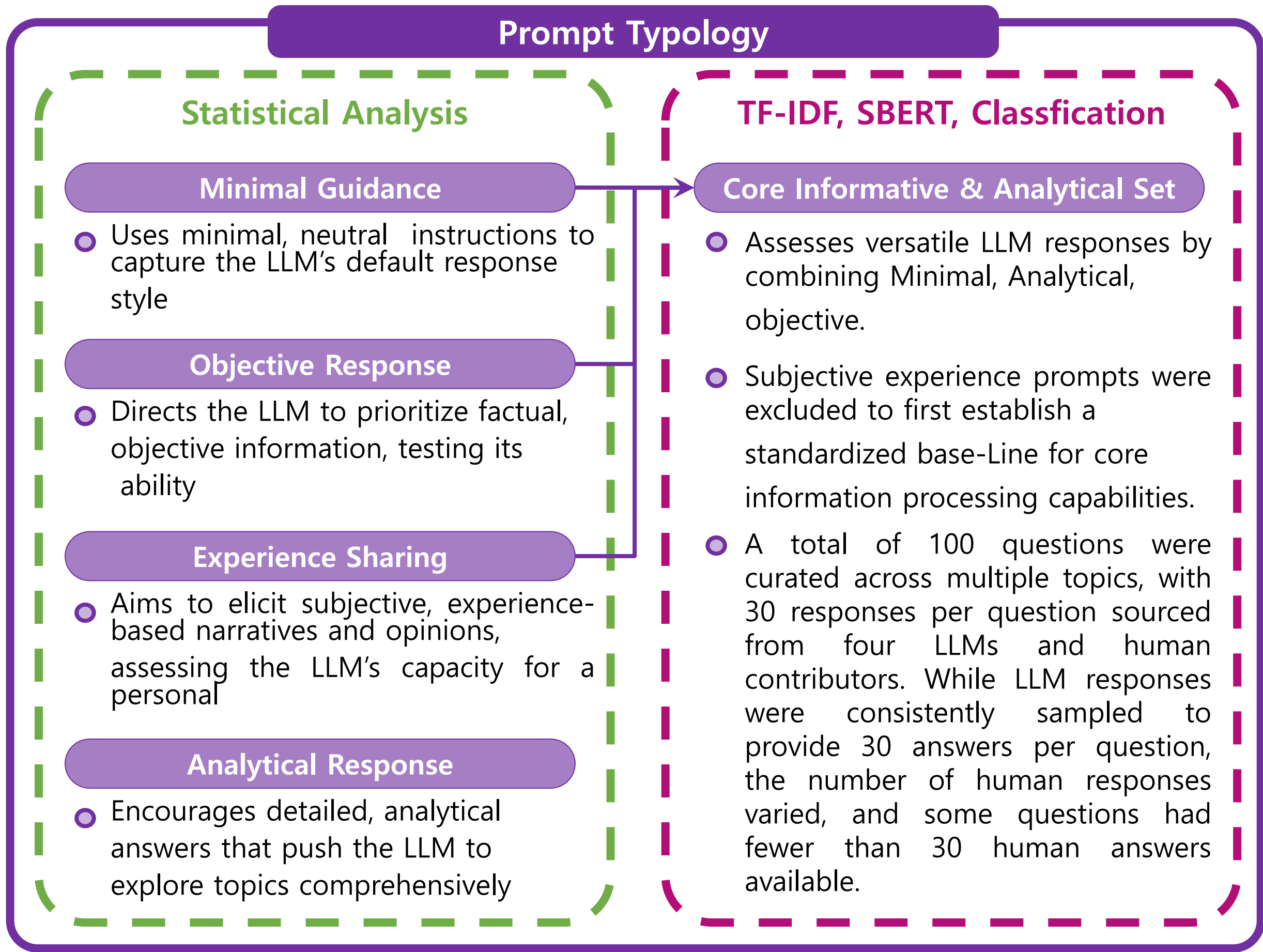
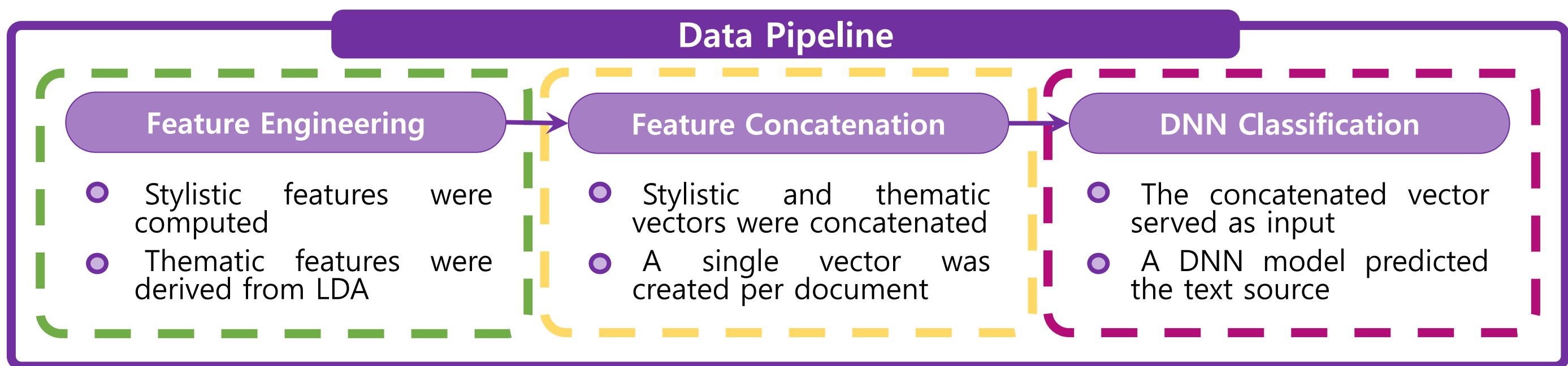
Instead of relying on a single signal, our method combines stylistic and readability features—such as sentence complexity and the use of personal pronouns—with thematic features derived from Latent Dirichlet Allocation (LDA), which capture what the text is about.

Data & Methodology

Our study began with the careful construction of a corpus. The dataset comprises a total of 14,832 documents, sourced from 5 distinct sources: four LLMs and human sources.

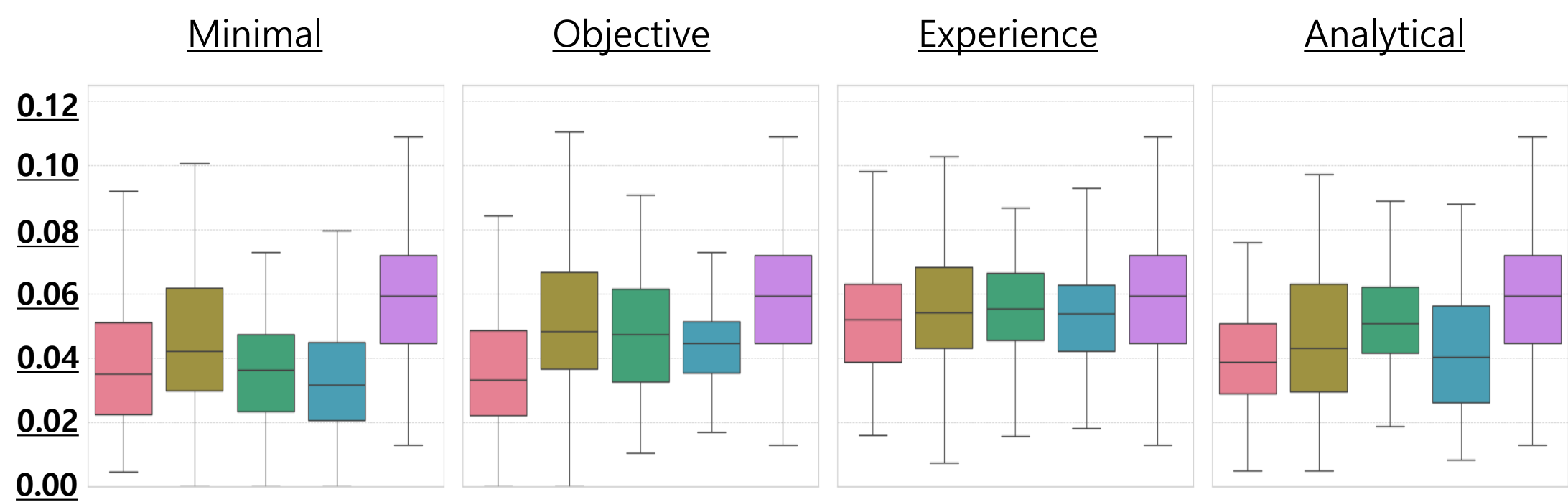
The entire corpus then underwent a standardized preprocessing pipeline, including lowercasing, tokenization, and the removal of stopwords and special characters.

For model selection, we tried three architectures (LightGBM, TabNet, Attention-based architecture). A Multi-Layer Perceptron (MLP) was ultimately selected for its highest performance on our dataset. The chosen network consists of four sequential hidden layers (256, 128, 64, 32), with Batch Normalization and Dropout applied after each layer. The preprocessed data was passed through the following three-step pipeline:

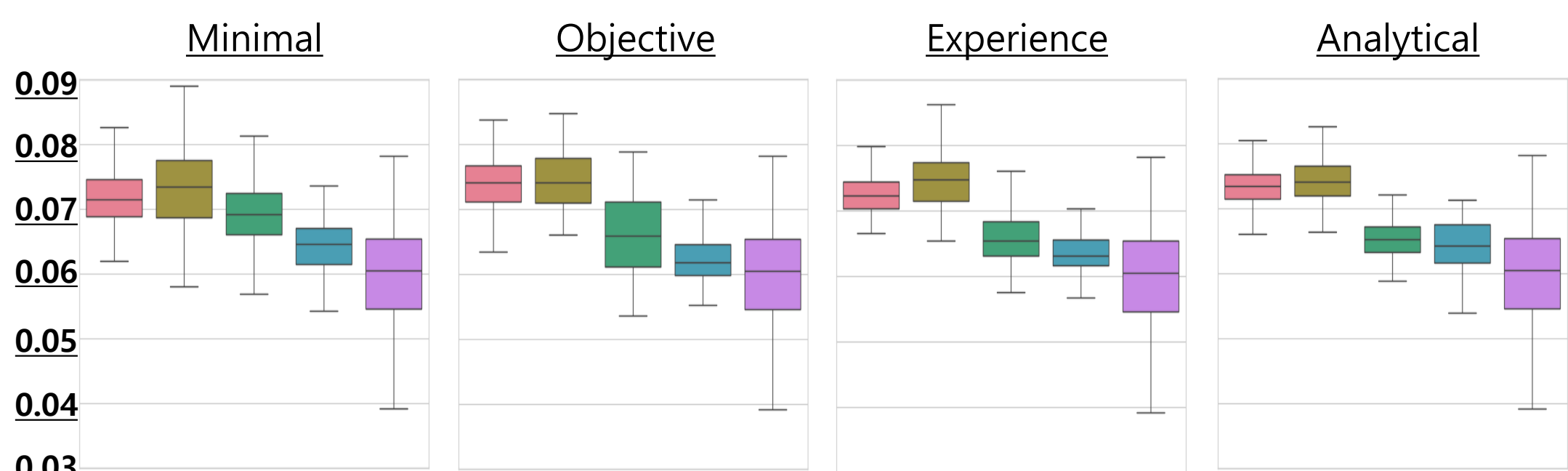


Syntactic Fingerprinting : How They Speak

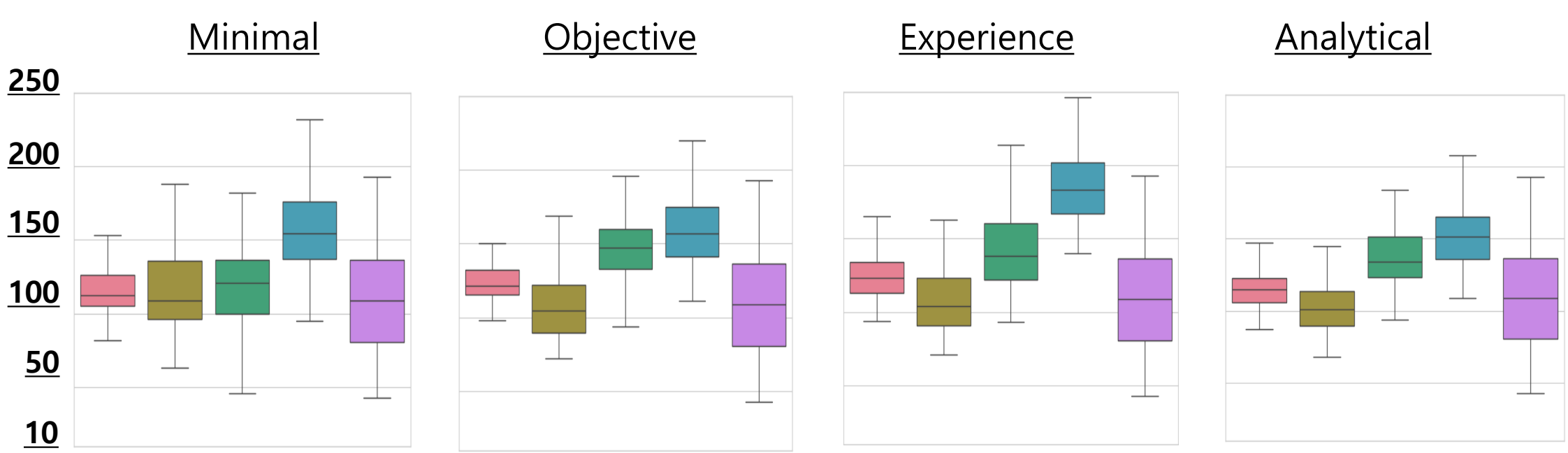
Personal Pronoun Proportion (PRP)



Lexical Density



Unique Word Count



Claude **DeepSeek** **Gemini** **GPT** **Human**

PRP Proportion = $\frac{\text{Number of Personal Pronouns}}{\text{Total Number of Words(Tokens)}}$

Lexical Density = $\frac{\text{Number of Unique Words}}{\text{Total Number of Words}}$

Topical & Lexical Signatures : What They Speak about

Term Frequency - Inverse Document Frequency (TF-IDF)

LLMs generate responses using a common, general-purpose lexicon. In contrast, human lexical choices indicate responses more grounded in specific contexts, personal perspectives, temporal awareness

Top-10	Human		GPT		Gemini		DeepSeek		Claude	
	Word	Score	Word	Score	Word	Score	Word	Score	Word	Score
1	One	0.2570	Like	0.3072	Like	0.3796	Like	0.3088	Like	0.2530
2	Time	0.2424	One	0.2373	Think	0.2711	One	0.2925	People	0.2247
3	Would	0.2235	Time	0.1840	Thing	0.2285	People	0.2696	I've	0.2189
4	People	0.1879	People	0.1837	One	0.1966	Even	0.2426	Think	0.2042
5	Like	0.1807	Think	0.1768	People	0.1805	Way	0.1815	Something	0.1804
6	Get	0.1582	Might	0.1735	Even	0.1754	Thing	0.1621	One	0.1750
7	Day	0.1550	Thing	0.1595	Okay	0.1751	Time	0.1506	Actually	0.1681
8	Year	0.1521	I've	0.1563	Really	0.1610	Make	0.1402	Thing	0.1529
9	Life	0.1457	Even	0.1517	Know	0.1607	Something	0.1389	Time	0.1517
10	dont	0.1381	Life	0.1473	something	0.1361	dont	0.1331	someone	0.1344

$$TF-IDF = TF * IDF$$

TF(Term Frequency) : How often a word appears in a document

IDF(Inverse Document Frequency) : How rare a word is across all documents

Classification of LLM and Human Text

LDA-Based Modeling

- Topic Modeling (LDA)
 - Construct Dictionary and Bag-of-Words from tokenized data
 - The number of topics was set to five (four LLMs and one human)
 - Represent each document as a distribution over main topics
- Subtopic Modeling
 - Select top 20 representative documents for each main topic
 - Identify optimal number of subtopics within each document group
 - Apply subtopic model to all documents to compute subtopic probability distributions
- A Deliberate Combination of Heterogeneous Features
 - Combine two fundamentally different feature categories
 - Integrate stylistic metrics (e.g., PRP) with thematic distributions from Hierarchical LDA
 - Create a richer feature set capturing a holistic view of text origin

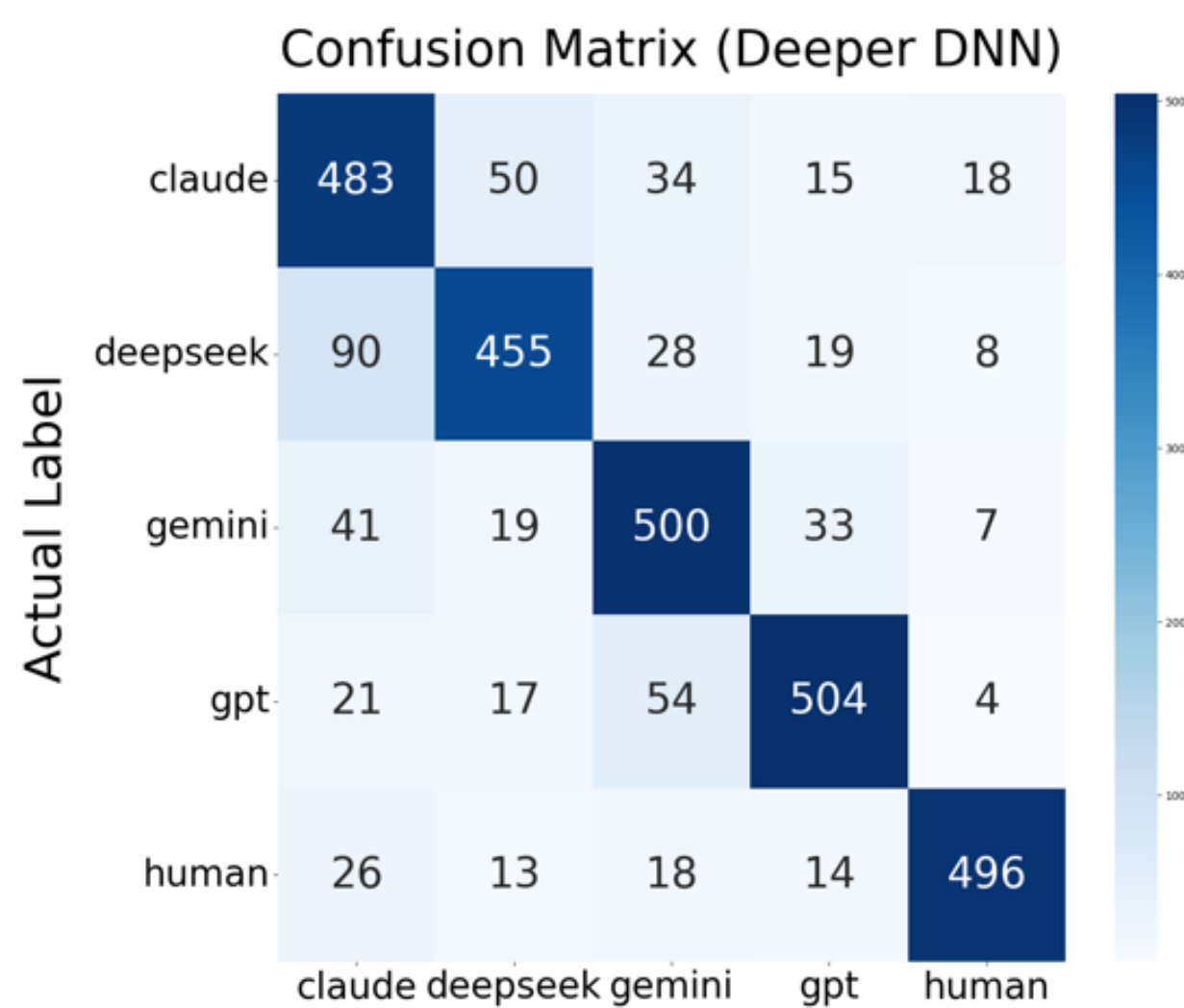
Doc_id	0-1	0-2	1-1	1-2	1-3	1-4	...	LLM_Name
1	0.7245	0.2755	0.1853	0.2493	0.2679	0.2975	...	GPT-4
2	0.9074	0.0926	0.2296	0.2947	0.1642	0.3115	...	Claude
3	0.8692	0.1308	0.2779	0.2186	0.2379	0.2656	...	Claude
4	0.8739	0.1261	0.1452	0.2068	0.3081	0.3399	...	Deepseek
...								
n-1	0.8486	0.1514	0.2454	0.2154	0.3099	0.2292	...	Human
n	0.8790	0.1210	0.1651	0.2923	0.2491	0.2935	...	Human

* This matrix is used as the input feature set for classification

DNN Based Classification

- Input X : Comprehensive hybrid features
- Target Y : LLM name or human associated with each document
- Classifier : DNN multi-class model (Splitting Train 80% and Test 20%)

	precision	recall	f1-score	support
claude	0.730711	0.805	0.7660586	600
deepseek	0.821299	0.758333	0.788561	600
gemini	0.788643	0.833	0.810372	600
gpt	0.861538	0.84	0.850632	600
human	0.930581	0.874779	0.901818	567
accuracy	0.821705			
Macro avg	0.826554	0.822289	0.823488	2967
Weighted avg	0.825397	0.821705	0.822617	2967



- The model had an overall accuracy of 82.2%, classifying human responses most accurately, while Claude and DeepSeek were often confused.
- The PRP ratio is particularly effective for discriminating between human and LLM texts, while lexical density shows its strength in classifying among the different LLM sources.
- LDA is helpful at all sources.

Additional: Ablation

- The model trained only on thematic (LDA) features achieved a notable 63.2% accuracy.
- The combination of thematic features and PRP achieves 78.9% accuracy.
- The use of thematic features, PRP, and lexical density achieves 81.2% accuracy

Conclusion & Future Work

This study successfully demonstrates a framework for classifying texts generated purely by either humans or LLMs. However, more features should be explored for the classifying LLM and human. Real world scenarios increasingly involve mixed-authorship texts (mixtexts), where human writing is revised by AI or vice versa. As highlighted by recent research (e.g., Zhang et al., 2024), detecting blended texts poses a significant challenge.

Acknowledgement

This research was supported by Global Learning & Academic research institution for Master's-PhD students, and Postdocs(LAMP) Program of the National Research Foundation of Korea(NRF) grant funded by the Ministry of Education (No. RS-2024-00443714).